

Analysis of Complex Sample Survey Data

SURV/SURVMETH 701

Lecture Notes: Module 7

Analysis Methods for Categorical Data

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Forms of Categorical Data

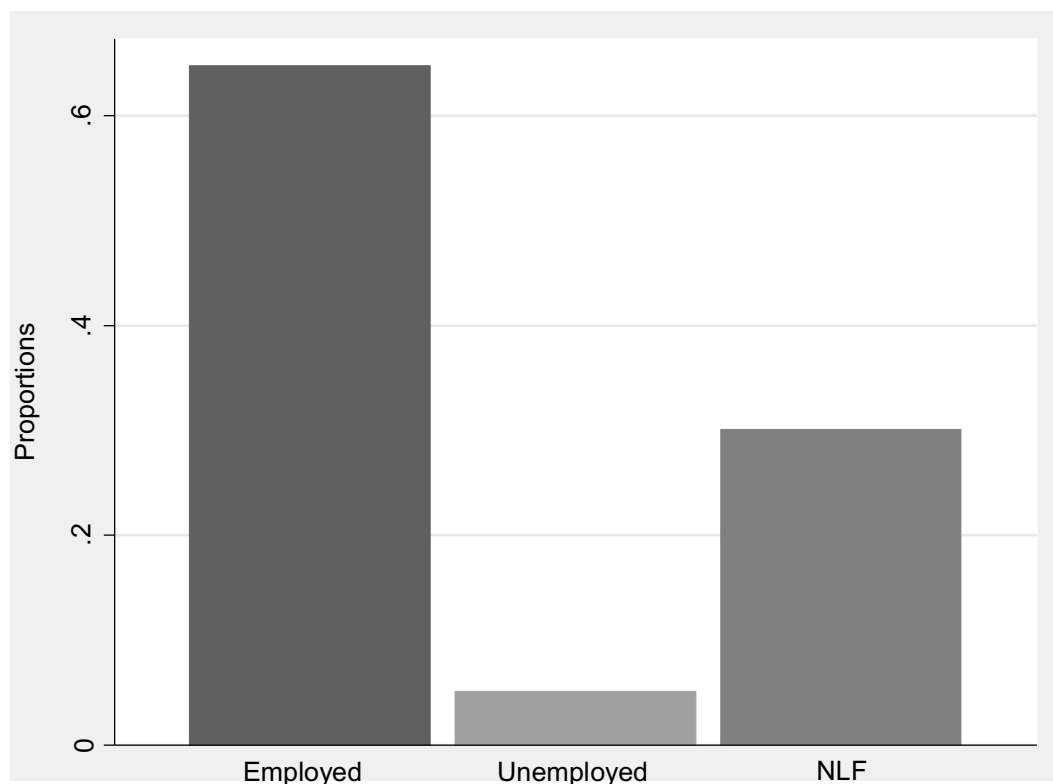
- Nominal - Categories are labels, no metric implied (e.g., Gender, Region).
- Ordinal - Categories imply ordered metric. e.g., education, grouped underlying continuous (age brackets).
- Counts – Special case of ordinal. Exact metric, discrete events, occurrences.

Nominal Categorical Data: NCS-R

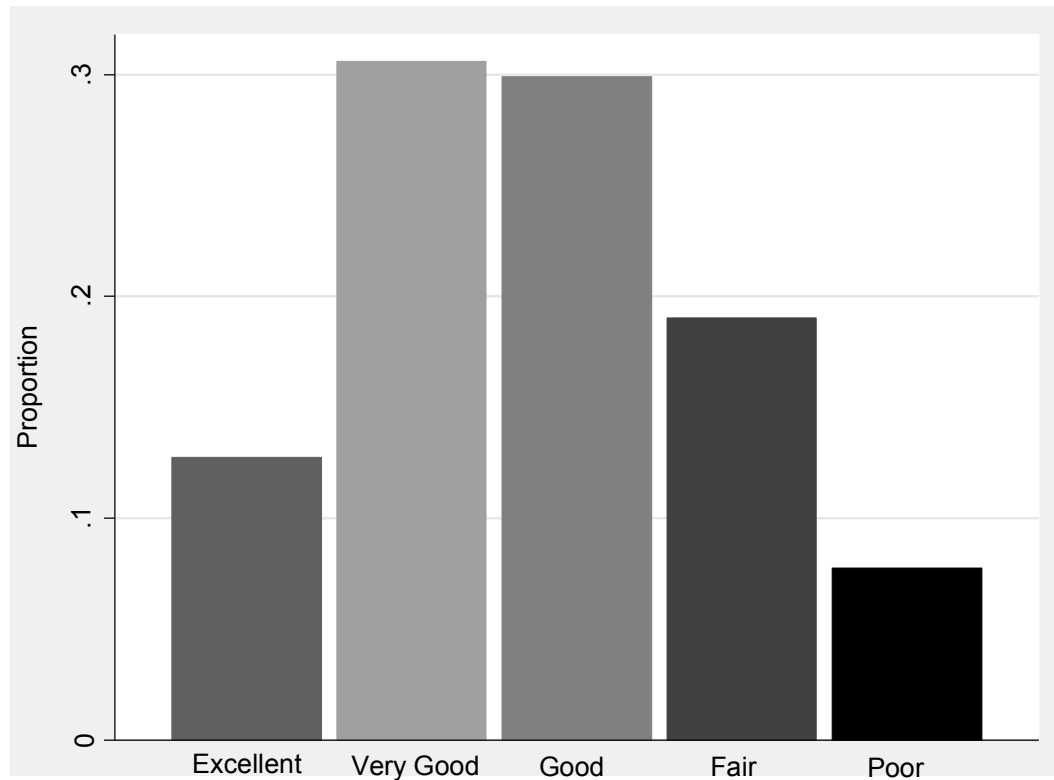
Weighted Distribution of U.S. Adults

by Employment Category

(NLF: not in the labor force)

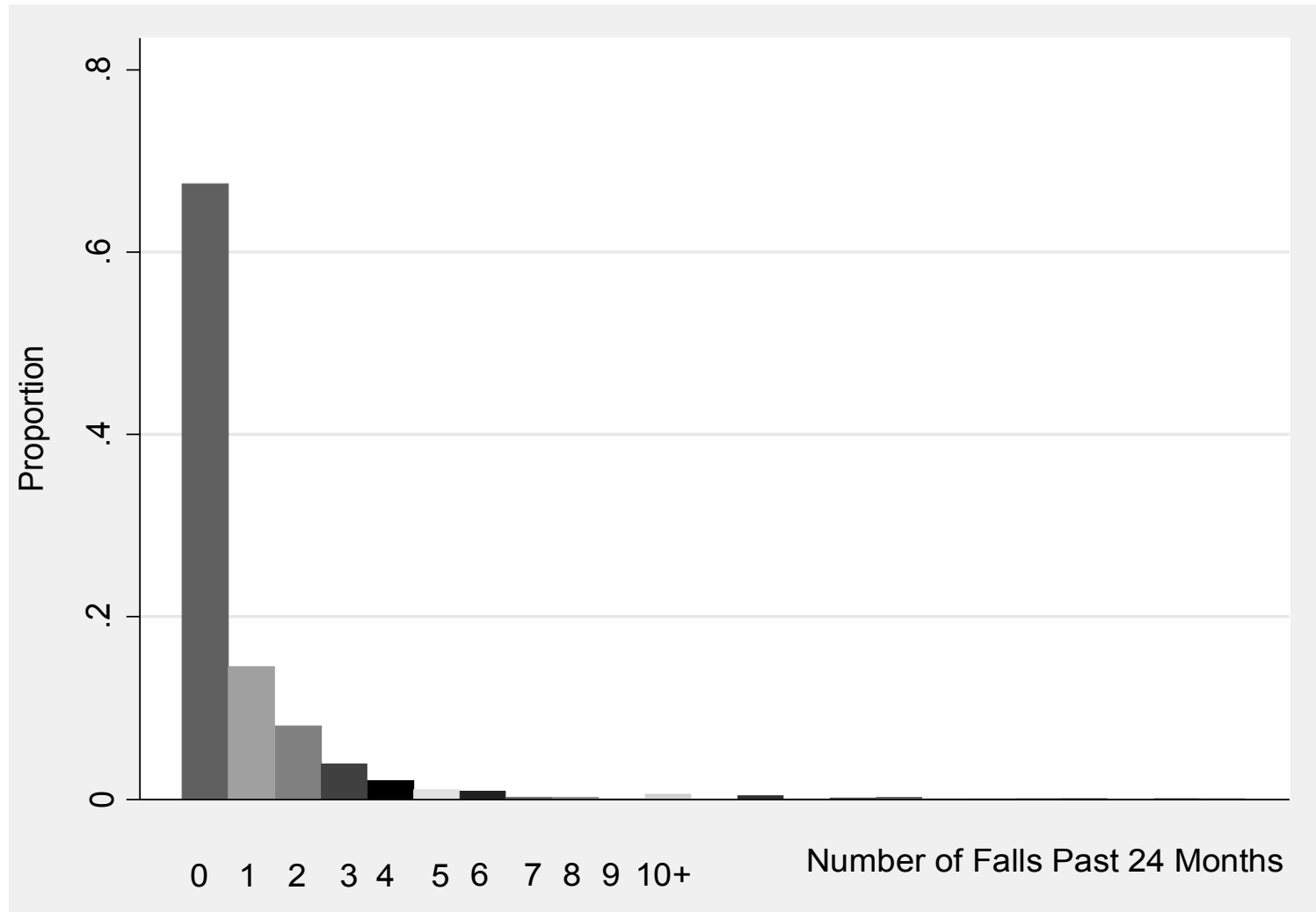


Ordinal Categorical Data: HRS 2006. Weighted Distribution. Self-rating of Health Status



Count Data: HRS 2006

Falls in the Past 24 Months



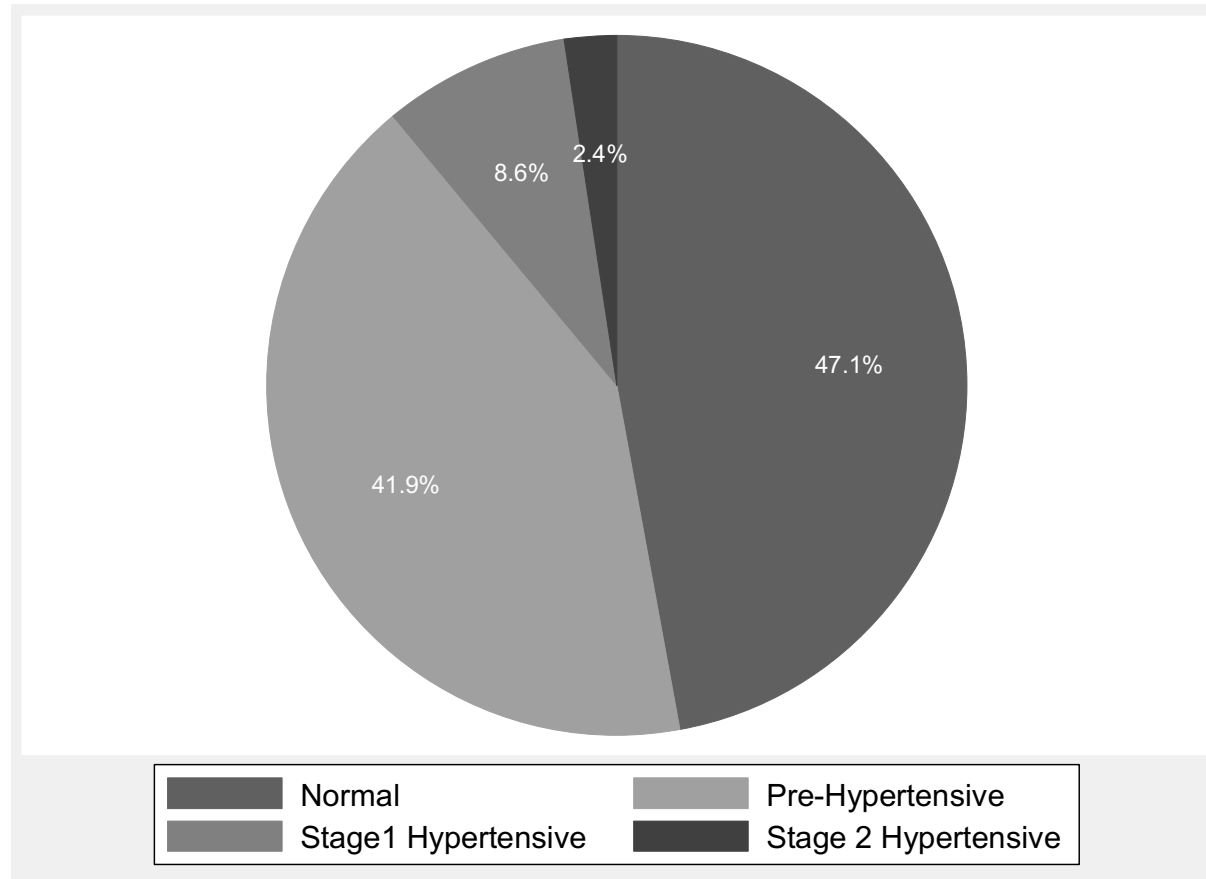
Categorical Data Analysis

- Univariate Tests (e.g., $H_0: p = p_0$)
- Bivariate and Multi-way Tables (here)
- Loglinear Models (not widely implemented in survey analysis software)
- Logistic Models (later in course)
- Poisson and other GLMs (later in course)

Univariate Methods

- Graphical display of data
- Primarily estimation of proportions and functions of proportions
- Ordinal, count – descriptive statistics for continuous variables used with caution
- Estimation of p , standard errors, CIs
- Goodness of fit tests

Estimated distribution:
U.S. Adults by Hypertension Status.
Source: NHANES



Univariate Analysis

- Estimation of simple proportions:

$$p = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{h\alpha}} w_{h\alpha i} I_{i \in 1}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{h\alpha}} w_{h\alpha i}} = \frac{\hat{N}_1}{\hat{N}} = \frac{y}{x}$$

- Recall that a proportion is simply the mean of $y = \{0,1\}$

Univariate Analysis

- TSL variance of simple proportions:

$$v(p) = \frac{V(\hat{N}_1) + p^2 V(\hat{N}) - 2p \text{Cov}(\hat{N}_1, \hat{N})}{\hat{N}^2}$$

- Recall: $\text{percent} = 100 \ p,$
 $\text{se}(\text{percent}) = 100 \ \text{se}(p),$
 $\text{var}(\text{percent}) = 100^2 \ \text{var}(p)$

Univariate Analysis

- Confidence intervals: asymptotic normal method (assume asymptotic normality of sampling distributions)
 - **Stata:** svy: mean, svy: ratio

$$CI_{.95}(\pi) = p \pm t_{.975,df} \cdot se(p)$$

Univariate Analysis

- Confidence intervals: logit transform method
 - **Stata:** svy: tab, svy: prop

$$CI[\text{logit}(\pi)] = \{A, B\} = \left\{ \begin{array}{l} \ln \frac{p}{1-p} - \frac{t_{1-\alpha/2, df} \cdot se(p)}{p \cdot (1-p)} \\ \ln \frac{p}{1-p} + \frac{t_{1-\alpha/2, df} \cdot se(p)}{p \cdot (1-p)} \end{array} \right\}$$
$$CI(\pi) = \left\{ \frac{e^A}{1 + e^A}, \frac{e^B}{1 + e^B} \right\}$$

Confidence Intervals for Proportions: Proportion 18+ with Irregular Heartbeat [NHANES (2011-12) – Recode not shown]

Stata Code:

```
svyset sdmvpsu [pweight = wtmecl2yr], strata(sdmvstra)
svy, subpop(age18p): tab irregular, se ci col deff
svy, subpop(age18p): proportion irregular
svy, subpop(age18p): mean irregular
```

Variable	<i>n</i>	Design <i>df</i>	Estimated Proportion	Linearized SE	95% CI	Design Effect
Stata svy: tab, CI based on logit transform technique						
Irregular	5,374	17	0.0164	0.0017	(0.0132, 0.0204)	1.21
Stata svy: prop, CI based on logit transform technique						
Irregular	5,374	17	0.0164	0.0017	(0.0132, 0.0204)	1.21
Stata svy: mean, CI computed using the standard symmetric interval						
Irregular	5,374	17	0.0164	0.0017	(0.0129, 0.0200)	1.21

Univariate Analysis: Multinomial Variables

- Variance estimation by TSL or Replication
- Stata: `svy: tab catvar, obs se`
- SAS: PROC SURVEYFREQ
- NHANES Example in SAS:

```
proc surveyfreq data = nhanes;  
weight wtmec2yr;  
stratum sdmvstra;  
cluster sdmvpsu;  
tables age18p*irregular / row cl; *Here age18p is a subpopulation;  
run;
```

Univariate Analysis: Multinomial

- Estimation follows that for binary proportions:

$$p_k = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{h\alpha}} w_{h\alpha i} I_k}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{i=1}^{n_{h\alpha}} w_{h\alpha i}} = \frac{\hat{N}_k}{\hat{N}}$$

Univariate Analysis: Multinomial

- Variance estimation by TSL or Replication
- CIs in Stata's `svy: tab` and `svy: prop` command use the logit transform method
- Goodness of Fit Hypothesis test ($H_0: \boldsymbol{\pi} = \boldsymbol{\pi}_0$):
Stata `mgof` command

Estimating Proportions for Multinomial Variables

- `svyset sdmvpsu [pweight = wtmecl2yr], strata(sdmvstra)`
- `svy, subpop(age18p): tab bp_cat, obs se ci col`
- `svy, subpop(age18p): tab bp_cat, deff`

Estimated Proportions of U.S. Adults by Blood Pressure Status. Source: 2011-2012 NHANES

Blood Pressure Category	<i>n</i>	Estimated Proportion	Linearized SE	95% CI	Design Effect
Normal	2438	0.472	0.016	(0.440, 0.505)	6.70
Pre-Hypertension	2284	0.428	0.012	(0.403, 0.454)	4.10
Stage 1 Hypertension	489	0.080	0.006	(0.068, 0.093)	3.19
Stage 2 Hypertension	145	0.020	0.004	(0.013, 0.03)	6.79

Example: Goodness of Fit for a Vector of Proportions: Hypothesized π for Hypertension Status Category

recode bp_cat (1=.5) (2=.3) (3=.15) (4=.05), generate(pi)

Next, the mgof command is submitted to test the null hypothesis:

mgof bp_cat = pi if age18p == 1, svy

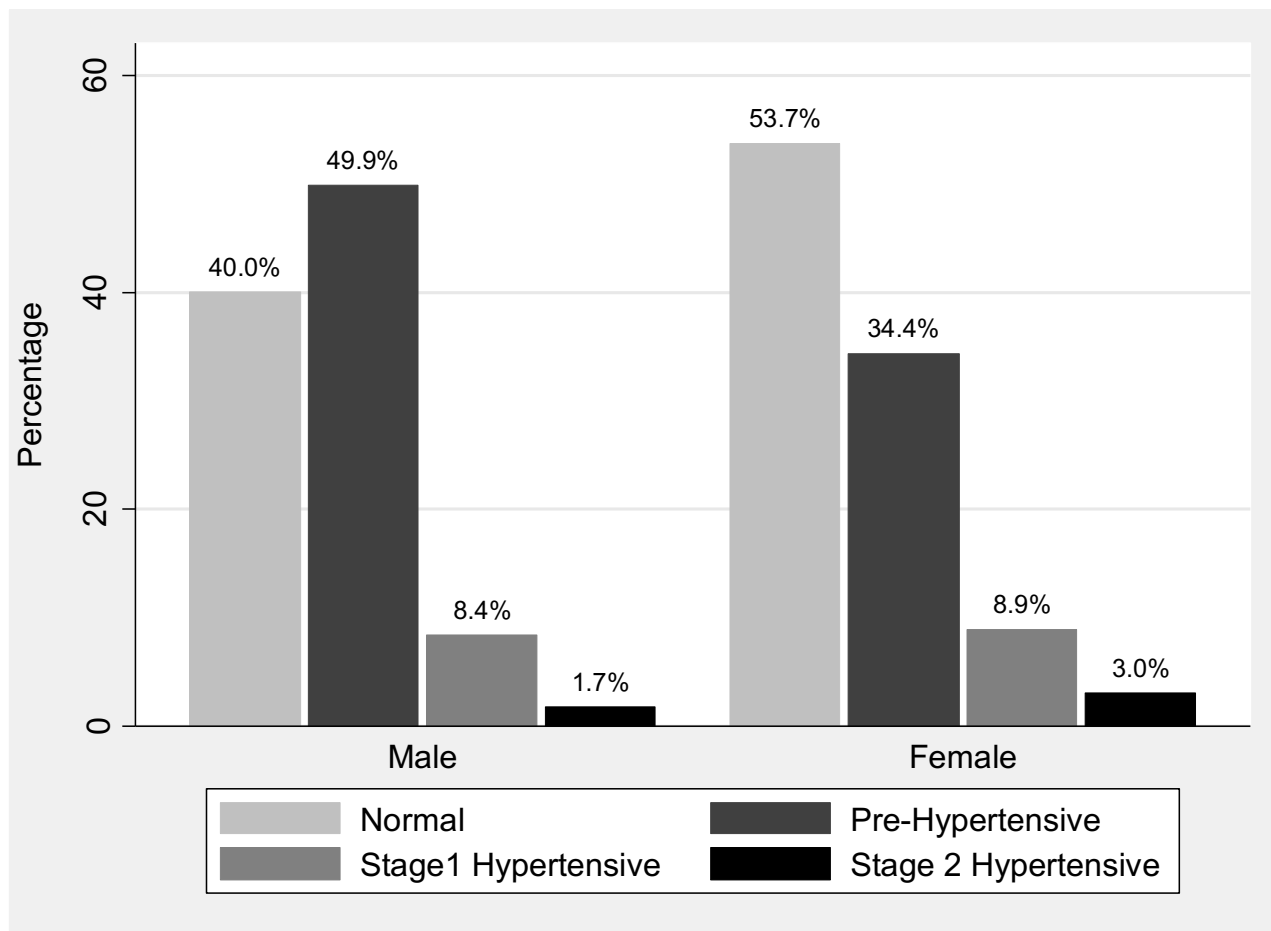
Goodness-of-fit Statistic	X^2	$p(\chi_3^2 > X^2)$
Rao-Scott X^2_{Pearson}	450.24	$p < 0.0001$
$X^2_{\text{LR, Jann}}$	465.48	$p < 0.0001$

Bivariate Categorical Data Analysis

- Contingency Tables
 - Total proportions
 - Row, column proportions
- Hypothesis Tests of Independence
- Odds Ratios, Relative Risk
- Simple Logit and Odds Ratios
- Regression Models (later)

Estimated distribution: Hypertension Status by Gender.

Source: NHANES



NCS-R Example: 2 x 2 Table

- Example: Suppose that we are interested in studying the association between gender and a lifetime major depression episode (1=yes, 0=no).

Proportions of Interest

Total proportion: e.g. $p_{M0} = \frac{\hat{N}_{M0}}{\hat{N}_{++}}$

Row proportion: e.g. $p_{0|M} = \frac{\hat{N}_{M0}}{\hat{N}_{M+}}$

Column proportion: e.g. $p_{M|0} = \frac{\hat{N}_{M0}}{\hat{N}_{+0}}$

Estimation of Total and Row Proportions

svyset seclustr [pweight = ncsrwts], strata(sestrat)

svy: tab sex mde, se ci deff

svy: tab sex mde, row se ci deff

Description	Parameter	Estimated Proportion	Linearized SE	95% CI	Design Effect
Total Proportions					
Male, no MDE	π_{A0}	0.406	0.007	(0.393, 0.421)	1.87
Male, MDE	π_{A1}	0.072	0.003	(0.066, 0.080)	1.64
Female, no MDE	π_{B0}	0.402	0.005	(0.391, 0.413)	1.11
Female, MDE	π_{B1}	0.120	0.003	(0.114, 0.126)	0.81
Row Proportions					
No MDE Male	$\pi_{0 A}$	0.849	0.008	(0.833, 0.864)	2.08
MDE Male	$\pi_{1 A}$	0.151	0.008	(0.136, 0.167)	2.08
No MDE Female	$\pi_{0 B}$	0.770	0.006	(0.759, 0.782)	0.87
MDE Female	$\pi_{1 B}$	0.230	0.006	(0.218, 0.241)	0.87

NCS-R: MDE x Gender

Weighted Estimates of Total Proportions

	MDE=NO (0)	MDE=YES (1)	Total
MEN (M)	$p_{A0}=0.406$	$p_{A1}=0.072$	$p_{A+}=0.478$
WOMEN (F)	$p_{B0}=0.402$	$p_{B1}=0.120$	$p_{B+}=0.522$
Total	$P_{+0}=0.808$	$P_{+1}=0.192$	$p_{++}=1.000$

NCS-R: MDE x Gender

Weighted Estimates of Row Proportions

	MDE=NO (0)	MDE=YES (1)	Total
MEN (M)	$P_{0 A} =$	$P_{1 A} =$	$p_{A+} =$
WOMEN (F)	$P_{0 B} =$	$P_{1 B} =$	$p_{B+} =$

Estimation of Total and Row Proportions (SAS)

```
proc surveyfreq data = ncsr;  
weight ncsrwtsh;  
stratum sestrat;  
cluster seclustr;  
tables sex*mde / row cl deff;  
run;
```

- The logistic or polytomous regression model framework can be used analyze categorical data when one variable can be designated as the dependent variable and other variables as explanatory or predictor variables.
- Many times we are interested only in investigating associations between a set of categorical variables.
- A common approach is to compute a chi-square statistic under the null hypothesis, and fail to reject the null hypothesis if this chi-square statistic is within the range of values that would be expected.

Simple Hypothesis Tests

- (2 x 2 table). Under the null hypothesis of independence:

i) as a Chi square "test of independence"

$$H_0 : P_{rc} = P_{r+} P_{+c}$$

ii) as a test of difference in two proportions,

$$H_0 : \Pr(\text{Depress} = c \mid \text{Sex} = r) = \Pr(\text{Depress} = c \mid \text{Sex} = r')$$

Chi-square Tests: Pearson and Likelihood Ratio

- The chi-square test statistic is a measure of distance between the expected and observed counts in cells.

- Pearson:

$$X_{Pearson}^2 = n_{++} \cdot \sum_r \sum_c (p_{rc} - \hat{\pi}_{rc})^2 / \hat{\pi}_{rc}$$

- Likelihood Ratio:

$$G^2 = 2 \cdot n_{++} \cdot \sum_r \sum_c p_{rc} \times \ln \left(\frac{p_{rc}}{\hat{\pi}_{rc}} \right)$$

- If the observed data are consistent with the null hypothesis, then we would expect this distance measure (the test statistic) to be small.

Adjusting for Design Effects

- How to incorporate the design features into this analysis? Since ignoring the design features (weighting) can introduce bias in the estimated proportions that goes into making the Chi-square statistics, the above analysis is not valid.
- Two approaches:
 - Fellegi (JASA, 1980) corrects the SRS Chi-square statistic.
 - Rao-Scott (1984, Biometrika), Rao-Thomas (1987)

Fellegi Method

- Compute the average design effect of the proportions involved in the Chi-square statistic. For the 4 proportions in the NCS 2 x 2 example in the handout example (different from text example), the average design effect is ~ 1.3575 .
- Divide the SRS Chi-square test statistic by this average design effect. The corrected Chi-square statistic is $108.2913 / 1.3575 = 87.64$.
- P-value is obtained by referring the corrected statistics to a Chi-square distribution with degrees of freedom $(r-1)(c-1)$.

Rao-Scott Method

- Use the weighted proportions in the construction of Chi-square statistics. Develop an approximate F-reference distribution for determining p-value.
- This is analytically more complicated and requires computation of generalized design effects. These are the eigenvalues of the matrix given on the next page.
- SAS, SUDAAN, and Stata have all implemented this approach.

Rao-Scott Generalized Design Effect

Generalized design effects are the eigen values of the matrix :

$$D = V_{Des} V_{SRS}^{-1}$$

where :

V_{Des} = *Design – based covariance matrix
for estimated proportions; and*

V_{SRS} = *SRS covariance matrix
for estimated proportions*

Design-adjusted X^2 Test Statistics

- First order correction (SAS system default):

$$X^2_{R\ S} = X^2_{Pearson} / GDEFF,$$

$$G^2_{R\ S} = G^2 / GDEFF$$

- *GDEFF: the mean of the eigenvalues, or the “average design effect” (see ASDA, p. 176)*

Design-adjusted X^2 Test Statistics

- Second order correction
- Stata system default, SAS V9.3 option:

$$X^2_{R\ S} = X^2_{Pearson} / GDEFF / (1 + a^2),$$

$$G^2_{R\ S} = G^2 / GDEFF / (1 + a^2)$$

where:

a = coefficient of variation of
eigenvalues of D matrix.

Rao-Scott Correction for 2-Way Tables

for a two – way table of proportions,

$$GDEFF = \left\{ \begin{array}{l} \sum_r \sum_c (1 - \hat{P}_{rc}) \cdot DEFF(\hat{P}_{rc}) \\ - \sum_r (1 - \hat{P}_{r.}) \cdot DEFF(\hat{P}_{r.}) \\ - \sum_c (1 - \hat{P}_{.c}) \cdot DEFF(\hat{P}_{.c}) \end{array} \right\} / (r-1)(c-1)$$

= Mean Generalized Design Effect

= Mean of eigenvalues of **D** matrix

Rao-Scott Chi-Square

(First order correction as in SAS default).

R – S Chi – Square:

$$Q_{rs} = Q_{\text{Pearson}} / GDEFF$$
$$\sim \chi^2_{(R-1)(C-1)} \text{ under } H_0$$

or :

$$F = Q_{rs} / (R-1)(C-1)$$
$$\sim F_{(R-1)(C-1), (R-1)(C-1)s} \text{ under } H_0$$

where : s = #clusters – #strata in complex design.

Rao-Scott Likelihood Ratio Chi-Square (First order correction as in SAS default).

$$R-S \text{ LR Chi-Square: } G^2_{RS} = G^2 / GDEFF$$

$$\sim \chi^2_{(R-1)(C-1)} \text{ under } H_0$$

$$\text{or: } F = G^2_{RS} / \{(C-1)(R-1)\}$$

$$\sim F_{(C-1)(R-1), s(C-1)(R-1)}$$

where: $s = \#clusters - \#strata$ in complex design.

Alcohol Dependence vs. Education Level for Young Adults Age 18-28. Source: NCS-R.

In Stata:

```
svyset seclustr [pweight = ncsrwtsh], strata(sestrat)  
svy, subpop(if 18<=age<29): tab ed4cat ald, row se ci deff
```

In SAS V9.2+:

```
title "Analysis Example : Proportions by Subgroups: NCSR" ;  
proc surveyfreq data=ncsr ;  
    strata sestrat ;  
    cluster seclustr ;  
    weight ncsrwtsh;  
    tables age29*ed4cat*ald / row deff chisq ;  
    * tables age29*ed4cat*ald / row deff chisq(secondorder) ; *(OPTIONAL);  
    format ed4cat edf. ald aldf. ; run ;
```

Table 6.8: Design-adjusted Analysis of Alcohol Dependence vs. Education Level for Young Adults Age 18-28. Source: NCS-R.

Education Level (Grades)	Alcohol Dependence (ALD) Row Percentages		
	0 = No	1 = Yes	Total
0-11	0.909 (0.029)	0.091 (0.029)	1.000
12	0.951 (0.014)	0.049 (0.014)	1.000
13-15	0.951 (0.010)	0.049 (0.010)	1.000
16+	0.931 (0.014)	0.069 (0.014)	1.000
Total	0.940 (0.009)	0.060 (0.009)	1.000
Tests of Independence (Note the difference!)			
Unadjusted X^2	$P (\chi^2_{(3)} > X^2_{\text{Pearson}})$	Rao-Scott F	$P (F_{2.75, 115.53} > F_{R-S})$
$X^2_{\text{Pearson}} = 27.21$	$p < 0.0001$	$F_{R-S, \text{Pearson}} = 1.64$	$p = 0.18$
Parameters of the Rao-Scott Design-Adjusted Test			
$n_{18-29} = 1275$	Design $df = 42$	$GDEFF = 6.62$	$a = 0.56$

Odds Ratios and Relative Risks

$$\begin{aligned}\hat{\psi} &= \frac{\text{Odds}(MDE = 1 \mid \text{Male})}{\text{Odds}(MDE = 1 \mid \text{Female})} = \frac{p_{1|A} / (1 - p_{1|A})}{p_{1|B} / (1 - p_{1|B})} = \frac{p_{1|A} / p_{0|A}}{p_{1|B} / p_{0|B}} \\ &= \frac{0.151 / 0.849}{0.230 / 0.770} = 0.595\end{aligned}$$

$$\begin{aligned}\hat{RR} &= \frac{\text{Prob}(MDE = 1 \mid \text{Male})}{\text{Prob}(MDE = 1 \mid \text{Female})} \\ &= \frac{p_{1|A}}{p_{1|B}} = \frac{0.151}{0.230} = 0.656\end{aligned}$$

Logit of a Proportion

$$\text{logit}(p_{1|A}) = \ln(\text{Odds}(MDE = 1 \mid \text{Male})) = \ln\left(\frac{p_{1|A}}{1 - p_{1|A}}\right)$$

$$= \ln\left(\frac{0.151}{0.849}\right) = -1.727$$

$$\text{logit}(p_{1|B}) = \ln(\text{Odds}(MDE = 1 \mid \text{Female})) = \ln\left(\frac{p_{1|B}}{1 - p_{1|B}}\right)$$

$$= \ln\left(\frac{0.230}{0.770}\right) = -1.208$$

Logit Regression and Odds Ratios

$$\text{logit}(P(MDE = 1 \mid SEX)) = \beta_0 + \beta_1 \cdot I_{male}$$

$$\begin{aligned}\hat{\psi} &= \frac{p_{1|A} / (1 - p_{1|A})}{p_{1|B} / (1 - p_{1|B})} = \frac{\exp(\text{logit}(p_{1|A}))}{\exp(\text{logit}(p_{1|B}))} \\ &= \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1)}{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 0)} = \exp(\hat{\beta}_1)\end{aligned}$$

**Confidence Intervals
for Unadjusted Odds Ratios**

$$CI(\psi) = \begin{pmatrix} CIL = \exp(\hat{\beta}_1 - t_{1-\alpha/2, df} \cdot se(\hat{\beta}_1)), \\ CIU = \exp(\hat{\beta}_1 + t_{1-\alpha/2, df} \cdot se(\hat{\beta}_1)) \end{pmatrix}$$